Exercises Set: Calculus

Solutions

DSBA Mathematics Refresher 2026

1 Fundamental Theorem of Calculus

Generalization / Corollary

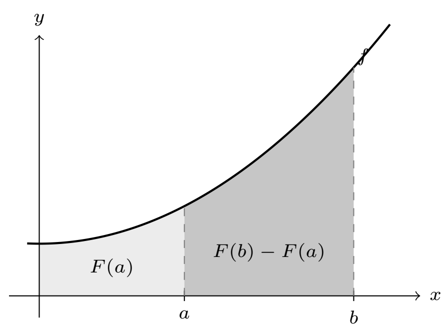
Let F be the particular anti-derivative $F(x) = \int_0^x f(t)\,dt$. Using the Chasles relation $\int_0^b = \int_0^a + \int_a^b$:

$$F(b) - F(a) = \int_0^b f(t)\,dt - \int_0^a f(t)\,dt = \left(\int_0^a f(t)\,dt + \int_a^b f(t)\,dt\right) - \int_0^a f(t)\,dt = \int_a^b f(t)\,dt$$

This proves the result for *that* anti-derivative. It holds in fact for **any** anti-derivative G of f: two anti-derivatives of the same function differ by a constant, so $G = F + C$, and the constant cancels in the difference:

$$G(b) - G(a) = (F(b) + C) - (F(a) + C) = F(b) - F(a) = \int_a^b f(t)\,dt$$

This is why we may pick whichever anti-derivative is most convenient, and why the constant of integration is irrelevant for *definite* integrals.



The light area is $\int_0^a f = F(a)$ and the dark area is $\int_a^b f$; together they make up $\int_0^b f = F(b)$. Subtracting the light area from the whole is exactly the statement $\int_a^b f = F(b) - F(a)$.

Application

An anti-derivative of sin is $-\cos$, so

$$\int_0^{\pi/2} \sin(x)\,dx = \Big[-\cos(x)\Big]_0^{\pi/2} = -\underbrace{\cos\left(\tfrac{\pi}{2}\right)}_{=0} + \underbrace{\cos(0)}_{=1} = 1$$

An anti-derivative of $\frac{1}{x^2} = x^{-2}$ is $-\frac{1}{x}$ (recall $\int x^n dx = \frac{x^{n+1}}{n+1}$ for $n \neq -1$, here with $n = -2$), so

$$\int_1^4 \frac{1}{x^2}\,dx = \left[-\frac{1}{x}\right]_1^4 = -\frac{1}{4} - \left(-\frac{1}{1}\right) = \frac{3}{4}$$

Note: both integrands are positive on the interval of integration, and both answers are positive — always worth a glance.

2 Integration Techniques

Substitution / Change of Variable

Exercise 1: $\displaystyle\int \frac{2x}{x^2+1}\,dx$

Let $u = x^2+1$; then $\dfrac{du}{dx} = 2x$, i.e. $du = 2x\,dx$, which is exactly the numerator:

$$\int \frac{2x}{x^2+1}\,dx = \int \frac{du}{u} = \ln|u| + C = \ln\left(x^2+1\right) + C$$

(no absolute value is needed since $x^2+1>0$).

Check: $\dfrac{d}{dx}\ln(x^2+1) = \dfrac{2x}{x^2+1}$. ✓

Remark: this is the pattern $\int \frac{g'(x)}{g(x)}dx = \ln|g(x)| + C$, which is worth recognising directly.

Exercise 2: (*) $\displaystyle\int \frac{1}{\sqrt{1-x^2}}\,dx$

Let $x = \sin(u)$, with $u \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ so that the substitution is invertible; then $dx = \cos(u)\,du$ and

$$\sqrt{1-x^2} = \sqrt{1-\sin^2(u)} = \sqrt{\cos^2(u)} = |\cos(u)| = \cos(u)$$

the last equality because $\cos(u) > 0$ on that interval. Therefore

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \int \frac{\cos(u)}{\cos(u)}\,du = \int 1\,du = u + C = \arcsin(x) + C$$

Check: this is the very definition of the derivative of arcsin. ✓

Exercise 3: $\displaystyle\int \frac{1}{4+x^2}\,dx$

Let $x = 2u$ (i.e. $u = x/2$); then $dx = 2\,du$ and $4+x^2 = 4+4u^2 = 4\left(1+u^2\right)$, so

$$\int \frac{1}{4+x^2}\,dx = \int \frac{2\,du}{4\,(1+u^2)} = \frac{1}{2}\int \frac{du}{1+u^2} = \frac{1}{2}\arctan(u) + C = \frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$$

Check: $\dfrac{d}{dx}\left[\frac{1}{2}\arctan\left(\frac{x}{2}\right)\right] = \frac{1}{2}\cdot\frac{1}{1+\frac{x^2}{4}}\cdot\frac{1}{2} = \frac{1}{4+x^2}$. ✓

Remark: the point of the substitution is to bring the integral back to the standard form $\int \frac{du}{1+u^2} = \arctan(u) + C$. The same trick gives $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C$.

Integration by Parts

Recall the formula, which comes from integrating the product rule $(uv)' = u'v + uv'$:

$$\int u\,v' = \Big[uv\Big] - \int u'\,v$$

The whole skill is choosing which factor plays the role of u (the one that gets *simpler* when differentiated).

Exercise A: $\displaystyle\int x\ln(x)\,dx$

Take $u = \ln(x)$ and $v' = x$ (so $u' = \frac{1}{x}$ and $v = \frac{x^2}{2}$) — differentiating the logarithm is what removes it:

$$\int x\ln(x)\,dx = \frac{x^2}{2}\ln(x) - \int \frac{x^2}{2}\cdot\frac{1}{x}\,dx = \frac{x^2}{2}\ln(x) - \frac{1}{2}\int x\,dx = \frac{x^2}{2}\ln(x) - \frac{x^2}{4} + C$$

that is

$$\int x\ln(x)\,dx = \frac{x^2}{2}\left(\ln(x) - \frac{1}{2}\right) + C$$

Check: $\dfrac{d}{dx}\left[\frac{x^2}{2}\ln x - \frac{x^2}{4}\right] = x\ln x + \frac{x^2}{2}\cdot\frac{1}{x} - \frac{x}{2} = x\ln x$. ✓

Exercise B: $\displaystyle\int x^2 e^x \, dx$

Take $u = x^2$ and $v' = e^x$: each integration by parts lowers the power of x by one, so we apply it twice.

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x \, e^x \, dx$$

and for the remaining integral, again with $u = 2x$ and $v' = e^x$:

$$\int 2x \, e^x \, dx = 2xe^x - \int 2e^x \, dx = 2xe^x - 2e^x$$

Therefore

$$\int x^2 e^x \, dx = x^2 e^x - 2xe^x + 2e^x + C = e^x \left(x^2 - 2x + 2\right) + C$$

Check: $\dfrac{d}{dx}\left[e^x(x^2 - 2x + 2)\right] = e^x(x^2 - 2x + 2) + e^x(2x - 2) = e^x x^2.$ ✓

Exercise C: (*) $\displaystyle\int x \cos(x) \, dx$

Take $u = x$ and $v' = \cos(x)$ (so $u' = 1$ and $v = \sin(x)$):

$$\int x \cos(x) \, dx = x \sin(x) - \int \sin(x) \, dx = x \sin(x) + \cos(x) + C$$

Check: $\dfrac{d}{dx}\left[x \sin x + \cos x\right] = \sin x + x \cos x - \sin x = x \cos x.$ ✓

Exercise D: $\displaystyle\int e^{2x} \cos(2x) \, dx$

Here neither factor simplifies when differentiated, so integrating by parts twice brings the *original* integral back — and we solve for it. Write

$$I = \int e^{2x} \cos(2x) \, dx \qquad \text{and} \qquad J = \int e^{2x} \sin(2x) \, dx$$

For I, take $u = \cos(2x)$ and $v' = e^{2x}$ (so $u' = -2\sin(2x)$ and $v = \frac{1}{2}e^{2x}$):

$$I = \frac{1}{2}e^{2x} \cos(2x) - \int \frac{1}{2}e^{2x} \left(-2\sin(2x)\right) dx = \frac{1}{2}e^{2x} \cos(2x) + J$$

For J, take $u = \sin(2x)$ and $v' = e^{2x}$ (so $u' = 2\cos(2x)$):

$$J = \frac{1}{2}e^{2x} \sin(2x) - \int \frac{1}{2}e^{2x} \cdot 2\cos(2x) \, dx = \frac{1}{2}e^{2x} \sin(2x) - I$$

Substituting the second relation into the first:

$$I = \frac{1}{2}e^{2x} \cos(2x) + \frac{1}{2}e^{2x} \sin(2x) - I \qquad \Longrightarrow \qquad 2I = \frac{1}{2}e^{2x} \left(\cos(2x) + \sin(2x)\right)$$

so

$$\int e^{2x} \cos(2x) \, dx = \frac{1}{4}e^{2x}\Big(\cos(2x) + \sin(2x)\Big) + C$$

Check: differentiating,

$$\frac{1}{4}\Big[2e^{2x} \left(\cos 2x + \sin 2x\right) + e^{2x} \left(-2\sin 2x + 2\cos 2x\right)\Big] = \frac{1}{4}e^{2x} \cdot 4\cos(2x) = e^{2x} \cos(2x) ✓$$

Further integration techniques

Exercise α: $\displaystyle\int \frac{3x^2-2x-1}{x^3-x^2+x-1}\,dx$

Factor the denominator by grouping:

$$x^3-x^2+x-1 = x^2(x-1)+(x-1) = (x-1)\left(x^2+1\right)$$

Factor the numerator: $x=1$ is an obvious root ($3-2-1=0$), so $(x-1)$ divides it, and

$$3x^2-2x-1 = (x-1)(3x+1)$$

The factor $(x-1)$ cancels, and for $x \neq 1$:

$$\frac{3x^2-2x-1}{x^3-x^2+x-1} = \frac{3x+1}{x^2+1}$$

We split this into the two standard pieces:

$$\int \frac{3x+1}{x^2+1}\,dx = \frac{3}{2}\int \frac{2x}{x^2+1}\,dx + \int \frac{1}{x^2+1}\,dx = \frac{3}{2}\ln\left(x^2+1\right) + \arctan(x) + C$$

valid on any interval not containing $x=1$.

Remark: had the numerator been $3x^2-2x+1$, it would have been *exactly* the derivative of the denominator, and the answer would simply have been

$$\int \frac{3x^2-2x+1}{x^3-x^2+x-1}\,dx = \ln\left|x^3-x^2+x-1\right| + C$$

Always compare the numerator with the derivative of the denominator before doing anything else.

Exercise β: (*) $\displaystyle\int_0^{+\infty} e^{-x}\,dx$

An improper integral is *defined* as a limit, so we integrate up to a and then let $a \to +\infty$:

$$\int_0^{+\infty} e^{-x}\,dx = \lim_{a\to+\infty}\int_0^a e^{-x}\,dx = \lim_{a\to+\infty}\Big[-e^{-x}\Big]_0^a = \lim_{a\to+\infty}\left(\underbrace{-e^{-a}}_{\longrightarrow 0} + \underbrace{e^0}_{=1}\right) = 1$$

The limit exists and is finite: the integral **converges**, and its value is 1.

Note: the area under an unbounded region can perfectly well be finite — here the exponential decays fast enough for the total area to be exactly 1. (This is the reason e^{-x} is a probability density on $[0,+\infty)$.)

Exercise γ: Trapezoidal rule, $n=3$, for $\displaystyle\int_0^{\pi/2} \sin(x)\,dx$

With $h = \dfrac{b-a}{n} = \dfrac{\pi/2}{3} = \dfrac{\pi}{6}$, the rule replaces the curve by a straight line on each sub-interval:

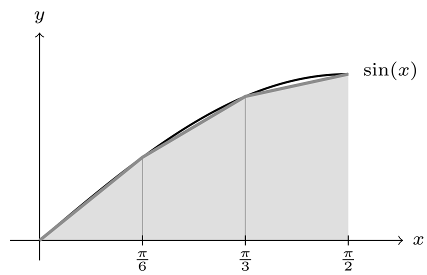
$$\int_a^b f(x)\,dx \approx h\left[\frac{1}{2}f(x_0) + f(x_1) + \cdots + f(x_{n-1}) + \frac{1}{2}f(x_n)\right]$$

(the interior points are counted once, the two endpoints only half each). The nodes are $0, \frac{\pi}{6}, \frac{\pi}{3}, \frac{\pi}{2}$, and

$$\int_0^{\pi/2} \sin(x)\,dx \approx \frac{\pi}{6}\left[\frac{\sin(0)}{2} + \sin\left(\frac{\pi}{6}\right) + \sin\left(\frac{\pi}{3}\right) + \frac{\sin\left(\frac{\pi}{2}\right)}{2}\right] = \frac{\pi}{6}\left[0 + \frac{1}{2} + \frac{\sqrt{3}}{2} + \frac{1}{2}\right]$$

$$= \frac{\pi}{6}\left(1 + \frac{\sqrt{3}}{2}\right) \approx 0.5236 \times 1.8660 \approx 0.977$$

The exact value is 1, so the error is about 2.3%.



The three trapezoids are bounded above by *chords* of the curve. Since sin is concave here, every chord lies *below* the arc it subtends, so the shaded area misses a small sliver under each arc.

The approximation is an *under*-estimate, which is expected: sin is concave on $[0, \pi/2]$, so every chord lies below the curve.

Exercise δ: Simpson's rule, $n = 2$, for $\displaystyle\int_0^{\pi/2} \sin(x)\, dx$

Simpson's rule replaces the curve by a *parabola* through three consecutive nodes:

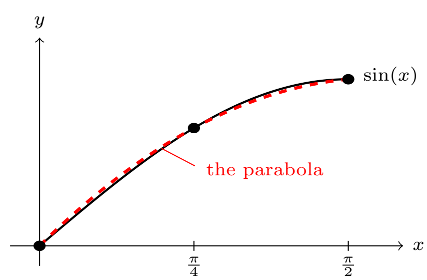
$$\int_a^b f(x)\, dx \approx \frac{h}{3}\left[f(x_0) + 4f(x_1) + f(x_2)\right], \qquad h = \frac{b-a}{n} = \frac{\pi/2}{2} = \frac{\pi}{4}$$

The nodes are $0, \frac{\pi}{4}, \frac{\pi}{2}$, and

$$\int_0^{\pi/2} \sin(x)\, dx \approx \frac{\pi}{12}\left[\sin(0) + 4\sin\left(\frac{\pi}{4}\right) + \sin\left(\frac{\pi}{2}\right)\right] = \frac{\pi}{12}\left[0 + 4 \cdot \frac{\sqrt{2}}{2} + 1\right] = \frac{\pi}{12}\left(2\sqrt{2} + 1\right)$$

$$\approx 0.2618 \times 3.8284 \approx 1.002$$

The error is now about 0.23% — ten times better than the trapezoidal rule with *more* nodes.



The dashed parabola through the three nodes is almost indistinguishable from sin on this interval — which is why three nodes here beat the four nodes of the trapezoidal rule.

A parabola simply fits a smooth curve much better than a straight line does.

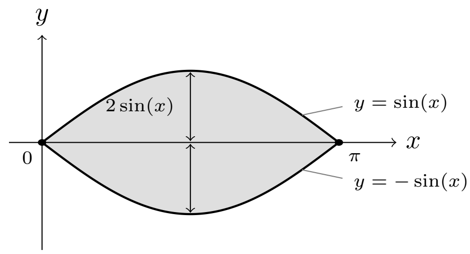
3 Applications

Areas between curves

The two curves $y = \sin(x)$ and $y = -\sin(x)$ meet where $\sin(x) = -\sin(x)$, i.e. $\sin(x) = 0$, which on $[0, \pi]$ happens at $x = 0$ and $x = \pi$ — the two ends of the interval. Between them $\sin(x) \geq 0$, so $\sin(x)$ is the upper curve and $-\sin(x)$ the lower one.

The area between two curves is the integral of (upper − lower):

$$\mathcal{A} = \int_0^{\pi} \Big(\sin(x) - (-\sin(x))\Big) dx = 2\int_0^{\pi} \sin(x)\, dx = 2\Big[-\cos(x)\Big]_0^{\pi} = 2\Big(\underbrace{-\cos(\pi)}_{=1} + \underbrace{\cos(0)}_{=1}\Big) = 4$$



The two dots are the points of intersection, and the vertical double arrow is the quantity being integrated: at each x, the region has height $\sin(x) - (-\sin(x)) = 2\sin(x)$.

The area is 4**.**

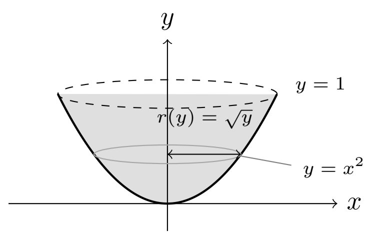
Note: the two curves are symmetric about the x-axis, so we could also have computed the area under sin on $[0, \pi]$ (which is 2) and doubled it.

Volume of a solid of revolution (*) (Disk Method)
We revolve about the y-axis, so we slice *horizontally*: at height y, the slice is a disk whose radius is the distance from the axis to the curve, i.e. the x such that $y = x^2$:

$$r(y) = \sqrt{y}$$

Each disk has area $\pi r(y)^2$ and thickness dy, so

$$V = \int_0^1 \pi\, r(y)^2\, dy = \int_0^1 \pi\,(\sqrt{y})^2\, dy = \pi \int_0^1 y\, dy = \pi \left[\frac{y^2}{2}\right]_0^1 = \frac{\pi}{2}$$



The grey ellipse is one of the disks being stacked: it sits at height y and has radius $\sqrt{y}$.

Sanity check: the solid sits inside the cylinder of radius 1 and height 1, whose volume is $\pi \approx 3.14$; we found $\frac{\pi}{2} \approx 1.57$, comfortably smaller. ✓

Arc length of curves
Here $f(x) = \frac{2}{3}x\sqrt{x} = \frac{2}{3}x^{3/2}$, so

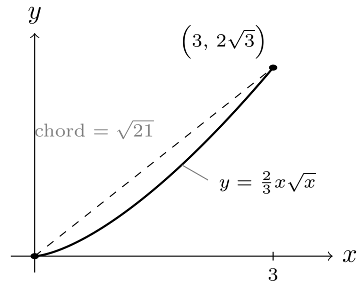
$$f'(x) = \frac{2}{3} \cdot \frac{3}{2} x^{1/2} = \sqrt{x} \qquad \text{and} \qquad 1 + \left(f'(x)\right)^2 = 1 + x$$

This is the point of the coefficient $\frac{2}{3}$: it makes the square root disappear from f', and the quantity under the arc length root becomes simply $1 + x$. Therefore

$$L = \int_0^3 \sqrt{1+x}\, dx = \left[\frac{2}{3}(1+x)^{3/2}\right]_0^3 = \frac{2}{3}\left(4^{3/2} - 1^{3/2}\right) = \frac{2}{3}(8-1) = \frac{14}{3} \approx 4.67$$

Check the anti-derivative: $\dfrac{d}{dx}\left[\frac{2}{3}(1+x)^{3/2}\right] = \frac{2}{3} \cdot \frac{3}{2}(1+x)^{1/2} = \sqrt{1+x}$. ✓

Sanity check: the curve joins $(0,0)$ to $\left(3, 2\sqrt{3}\right) \approx (3, 3.46)$, and the straight segment between these two points has length $\sqrt{3^2 + \left(2\sqrt{3}\right)^2} = \sqrt{21} \approx 4.58$. An arc is always at least as long as its chord, and $\frac{14}{3} \approx 4.67$ is indeed slightly longer. ✓



The arc (solid) and its chord (dashed) are visibly close, which is exactly what the sanity check expressed numerically.

Remark: do not be fooled by how smoothly this one went. Arc length integrals are *usually* horrible: $\sqrt{1+(f'(x))^2}$ rarely has an elementary anti-derivative — already for the parabola $f(x)=x^2$ it does not. The exercises that can be done by hand are precisely those where $1+(f')^2$ happens to be a perfect square or a simple power.

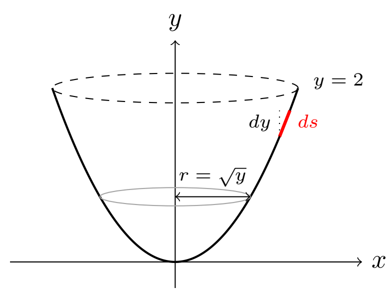
Surface area of a solid of revolution

We revolve the curve $y=x^2$, for $y\in[0,2]$, about the y-axis.

The surface area of a solid of revolution is *not* $\int 2\pi r\,dy$: a strip of the surface is slanted, so its width is the arc length element ds, not dy. The correct formula is

$$\mathcal{A}=\int 2\pi\, r\, ds$$

where r is the distance to the axis of revolution.



The picture shows why: the strip of surface (in red) follows the *slant* of the curve, so it is longer than the vertical rise dy (dotted).

Since we revolve about the y-axis, it is easiest to integrate with respect to y. On the curve, $x=\sqrt{y}$, so the radius is $r=\sqrt{y}$, and

$$\frac{dx}{dy}=\frac{1}{2\sqrt{y}} \qquad\Longrightarrow\qquad ds=\sqrt{1+\left(\frac{dx}{dy}\right)^2}\,dy=\sqrt{1+\frac{1}{4y}}\,dy=\sqrt{\frac{4y+1}{4y}}\,dy=\frac{\sqrt{4y+1}}{2\sqrt{y}}\,dy$$

The two $\sqrt{y}$ now cancel, which is what makes this computation short:

$$2\pi\, r\, ds=2\pi\sqrt{y}\cdot\frac{\sqrt{4y+1}}{2\sqrt{y}}\,dy=\pi\sqrt{4y+1}\,dy$$

so

$$\mathcal{A}=\pi\int_0^2\sqrt{4y+1}\,dy=\pi\left[\frac{1}{6}\,(4y+1)^{3/2}\right]_0^2=\frac{\pi}{6}\left(9^{3/2}-1\right)=\frac{\pi}{6}\,(27-1)=\frac{13\pi}{3}\approx 13.61$$

Check the anti-derivative: $\dfrac{d}{dy}\left[\frac{1}{6}(4y+1)^{3/2}\right]=\frac{1}{6}\cdot\frac{3}{2}\cdot 4\cdot(4y+1)^{1/2}=\sqrt{4y+1}.$ ✓

Cross-check, integrating in x instead: the interval $y\in[0,2]$ corresponds to $x\in\left[0,\sqrt{2}\right]$, the radius is $r=x$ and $ds=\sqrt{1+4x^2}\,dx$, so

$$\mathcal{A}=\int_0^{\sqrt{2}}2\pi x\sqrt{1+4x^2}\,dx=2\pi\int_1^9\sqrt{w}\,\frac{dw}{8}=\frac{\pi}{4}\left[\frac{2}{3}w^{3/2}\right]_1^9=\frac{\pi}{6}\,(27-1)=\frac{13\pi}{3}\ ✓$$

(with $w=1+4x^2$, $dw=8x\,dx$). Same answer, as it must be — but notice how much more pleasant the y version was.